Locate, Then Load

Foundations of a Unified Theory of
Capacity-Constrained Service under Imperfect Locatability

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Abstract

A class of economic systems is governed by a single invariant. In each of them, a single act of service—a *departure*—incurs a fixed cost against a finite capacity, and the units it serves live at *endpoints* whose location is not free to know. Two things therefore decide whether any departure is worth undertaking: how *locatable* its endpoints are, and how *full* the departure is. This paper states, in the most general form we can defend, the three primitives that follow—the *rate primitive* (locatability sets the service rate), the *law of consolidation* (fixed cost over finite capacity sets a load-factor threshold), and the *flywheel* (contribution feeds back into locatability, producing two stable equilibria separated by a separatrix)—together with the principle that decides which of the system's assets endure: the durable asset is the event stream, not the coordinate. The resulting theory is universal in form and local in parameters. Logistics, public transit, freight, cloud computing, hospital scheduling, and university enrolment are instances of it, differing only in the values of five parameters. The theory is falsifiable: we state the observations that would refute each of its theorems.

1 The Domain and Its Primitives

The objects below are deliberately abstract. They name a structure, not an industry. Each concrete system that satisfies the five scope conditions of §7 is an instance of them.

Definition 1 (Unit)**.** *A* unit *is an indivisible item of service demand: a parcel, a seat, a tonne, a procedure, a patient, a compute job, a guest, a student. A unit has a* value p *to the party it serves and a* variable cost c_v *of serving it.*

Definition 2 (Endpoint and locatability)**.** *An* endpoint *is the locus at which a unit is realised—a handover, an arrival, an admission. Its* locatability *is a number* $a \in (0, 1]$*, the degree to which the endpoint can be found, accessed, and served without negotiation.* $a = 1$*: the endpoint is coded, documented, and predictable.* $a \to 0$*: it exists only in someone's memory and is renegotiated at every visit.*

Definition 3 (Departure)**.** *A* departure *is a single committed act of service. It bears a* fixed cost c_0 *(crew, fuel, facility, dispatch, setup) that is paid whether it serves one unit or its full* capacity K*, and it carries* $N \le K$ *units. Its* load factor *is*

$$\lambda = \frac{N}{K} \in [0, 1].$$

Definition 4 (Bucket)**.** *A* bucket *is the object that exists* before *a departure is committed. It is the quadruple* $\mathcal{B} = (K, c_0, \tau, \lambda^*)$*, where* K *is capacity,* c_0 *the fixed cost incurred at commitment,* τ *a cutoff time, and* λ^* *the kill threshold below which the departure must not be fired. It accumulates committed units* $n(t)$ *for* $t < \tau$*, and then either fires or is killed.*

Definition 5 (Accumulation and the ledger)**.** *A system* accumulates *when each realised unit appends an event to a record associated with its endpoint. The record is a* ledger*. Accumulation is what turns today's service into tomorrow's knowledge.*

2 The Axiom and the Rate Primitive

Axiom 1 (Locatability). *A departure over an endpoint is economically calculable only if the endpoint is locatable. The cost of not locating it is not an overhead but a marginal cost, borne at the moment of service, and it scales with the opportunity cost of whatever waits at the unresolved endpoint:*

$$\lambda_r = w_{asset} \cdot \sigma,$$

where w_{asset} is the per-unit-time opportunity cost of the capacity idling at the endpoint, and σ is the characteristic duration of a search or negotiation.

The axiom implies that locatability is not a background condition. It is a first-order input to cost, and it operates through a single channel: the *rate* at which endpoints can be served.

Definition 6 (First-attempt success). *Let $p(a)$ be the probability a unit is realised on the first attempt. It is increasing and concave in a:*

$$p(a) = p_{\min} + (1 - p_{\min})\, a^{\gamma}, \qquad p_{\min} > 0,\ \gamma > 0.$$

Definition 7 (Service time and the rate). *Expected service time per unit at locatability a is*

$$\mathbb{E}[\tau(a)] = \tau_{load} + \tau_{travel} + \lambda_r \frac{1-a}{a} + \big(1 - p(a)\big)\, \tau_{reattempt},$$

and the service rate *is*

$$R(a) = \frac{1}{\mathbb{E}[\tau(a)]}.$$

Proposition 1 (The rate is the primitive). *Every short-run quantity of interest—cost per unit, units per departure, capacity per unit time, and the feasibility of a guaranteed service window—is monotone in $R(a)$. Locatability enters the short-run problem* only *through $R(a)$ and through the variance of τ.*

At low a, two facts hold simultaneously and multiply rather than add. The service time $\tau(a)$ grows—so fewer units fit in a fixed horizon—and the failure term $(1 - p(a))\, \tau_{\text{reattempt}}$ grows—so a share of each horizon is spent on attempts that produce nothing. A unit of locatability is therefore worth more than a unit of any other input, because it lowers cost and raises throughput at once.

3 The Law of Consolidation

The single most important structural fact in this class of systems is that the fixed cost is paid against the whole departure, while the margin is earned at the unit.

Law 1 (The Consolidation Invariant). *For a departure carrying N units at margin $p - c_v$ per unit and fixed cost c_0:*

$$G(N) = N\,(p - c_v) - c_0,$$

and the departure is profitable if and only if it crosses the break-even load factor

$$\boxed{\lambda > \lambda^* = \frac{c_0}{K\,(p - c_v)}} \qquad \textit{equivalently} \qquad N > N^* = \frac{c_0}{p - c_v}.$$

Corollary 1 (The marginal unit is pure margin; the empty slot is pure loss). *Below capacity, the marginal unit contributes the full margin $p - c_v$ with no share of c_0 allocated to it. Every unused unit of capacity $K - N$ is margin bought by c_0 and never recovered. The two facts jointly define the discipline of filling: the fixed cost is paid whether the departure is full or empty, the next unit is free, and the next empty slot is a realised loss of c_0/K.*

Corollary 2 (Capacity is a choice, and it has a ceiling)**.** *For a corridor of demand density, there exists an optimal capacity K^* that balances the falling fixed cost per unit against the falling load factor; K^* rises with density, and choosing K larger than the corridor can fill is a subsidy. In the full theory (§8) the choice is further bounded above by physical infrastructure:*

$$\lambda^*(I) = \frac{c_0}{K_{\max}(I)\,\big[\,p - c_v(I)\,\big]},$$

so that infrastructure is a capacity ceiling*, not a background condition.*

4 The Decomposition and the Durability Principle

Locatability is not one thing. It is two things wearing the same coat, and the distinction is the hinge of strategy.

Law 2 (The Static–Dynamic Decomposition)**.** *Locatability decomposes additively:*

$$a_j = a_j^S + a_j^D(n_j),$$

where the static *component a^S (the coordinate, the landmark, the access point) is purchasable in advance by survey, and the* dynamic *component a^D (reachability, first-attempt history, dwell-time stability) is earnable only through realised service. Formally,*

$$\frac{\partial a^S}{\partial n_j} = 0, \qquad \frac{\partial a^D}{\partial n_j} > 0.$$

Law 3 (The Durability Principle)**.** ***In any system satisfying the scope conditions of §7, the durable strategic asset is the event stream, not the coordinate.*** *Static locatability acquired at cost is a means of entry—a* bootstrap*—not a moat. The dynamic component, which can only be accumulated by operating, is the only asset that survives the arrival of whatever public or substitute supply eventually serves the static component for free.*

The Durability Principle has a plain form: *that which a public authority can give away is not defensible; that which requires operating is.* It converts the accumulation of §1 from an accounting detail into the central strategic fact of the theory.

5 The Flywheel

So far the chain is linear: locatability sets the rate, the rate sets cost, cost sets contribution. But contribution is not the end of the chain. It is the input to the next iteration, because contribution funds the service whose events accumulate locatability.

Theorem 1 (The Flywheel and Bistability)**.** *Let $\phi = \partial G/\partial a$ be the sensitivity of contribution to locatability, and $\psi = \partial a/\partial G$ the sensitivity of locatability to contribution through the ledger, so that the* loop gain *is $\Lambda = \phi\,\psi$. If $\Lambda > 1$, the system has two stable equilibria separated by a separatrix: a* virtuous cycle *in which service funds the accumulation that makes service cheaper, and a* death spiral *in which losses dilute service and starve the accumulation that would reverse them. The separatrix coincides with the break-even threshold λ^* of Law 1.*

Sketch. The map $a \mapsto a + \phi\psi\,(a - a_{\text{sep}})$ is expanding away from the separatrix. On either side, diminishing returns (negative second-order terms in ϕ and ψ) bound the motion into an attractor. The static threshold of Law 1 and the dynamical boundary coincide because both are the equation $G = 0$ in the same variables. □

Corollary 3 (Capital does not rescue the wrong side)**.** *Below the separatrix, an infusion of capital accelerates the spiral: it buys volume that dilutes the departures further. Capital is a positive perturbation only on the right side of the separatrix. This is why capital cannot save a structurally loss-making position, and why the previous generation of funded platforms failed at scale rather than despite it.*

6 The Bucket as a Bulk-Service Queue

The bucket of Definition 5 is not an analogy for a queue. It *is* one, and the identification transfers a mature literature to the theory.

Theorem 2 (The Bucket–Queue Equivalence)**.** *Every bucket $\mathcal{B} = (K, c_0, \tau, \lambda^*)$ is a bulk-service queue with maximum batch size K, minimum batch size $\lceil \lambda^* K \rceil$, service cost c_0, threshold ("N-policy") $Q = \lceil \lambda^* K \rceil$, and finite buffer with abandonment at τ. Conversely, every bulk-service queue with a service cost and a finite buffer is a bucket.*

The known algorithms therefore transfer directly. The kill rule is the *N-policy* (serve when the queue reaches the critical batch size). Route continuation after dispatch is the *(a, c, d)-policy* (a lower threshold once the startup cost is sunk). The cutoff decision is an *SMDP control-limit* policy. Admission control across heterogeneous units is *yield management*—the airline seat-allocation problem is, formally, optimal admission to a bulk-service queue (Subramanian, Stidham, and Lautenbacher state this equivalence explicitly). The theory need not invent its mathematics; it inherits it.

7 Scope and Universality

Definition 8 (The five scope conditions)**.** *The theory applies to a system if and only if all five hold simultaneously:*

1. ***Imperfect locatability.*** *A material share of endpoints have $a < 1$.*
2. ***Search cost that scales with the asset.*** *$\lambda_r = w_{asset}\sigma > 0$, rising with the opportunity cost of the capacity idling at the unresolved endpoint.*
3. ***A partially informal supply base.*** *A share $1 - \kappa > 0$ of capacity suppliers is not registered for consumption tax, so a firm buying and reselling capacity faces a principal burden $\Phi = v(1 - \kappa) + r\Delta t > 0$.*
4. ***Slow public supply of the static component.*** *Public provision of a^S arrives, but at a bounded rate θ, leaving a finite window in which private acquisition of a^S is defensible.*
5. ***Non-degenerate endpoint variance.*** *The same endpoint behaves differently across visits, so a^D is real rather than decorative.*

Where the conditions hold, the theorems hold. Where they fail, the theory is silent. This is what makes it a theory rather than a slogan: its scope is stated precisely enough to be wrong in a detectable way.

The class is broad. It contains the informal settlements of Asia, Africa, and Latin America; the rural regions of developed economies; refugee and disaster-response logistics; cross-border freight at uncoded customs points; and any dense urban market for high-value goods whose recipients cannot be predicted. It contains, at their own parameter values:

System	K	c_0	λ^*	The endpoint
Boda-boda (motorcycle)	1	low	degenerate	passenger at a stage
Informal minibus (matatu)	14	low	≈ 0.63	stage
City bus	60	medium	≈ 0.93	stop on a schedule
Long-haul freight truck	$\sim$1 load	high	high	yard / depot
Passenger train	1300	very high	0.10–0.33	station
Freight train	3000 t	very high	≈ 0.10	terminal
Aircraft	$\sim$180 seats	very high	0.75–0.85	airport
Cloud instance	reserved slots	provisioning	varies	machine in a datacenter
Operating theatre	theatre hours	team + theatre	varies	the patient
University class	seats	instructor + room	varies	the enrollee
Restaurant	tables	kitchen + staff	varies	the guest

8 The Ceiling and the Substitution Boundary

Two results bound the theory's reach and make it honest.

Theorem 3 (Infrastructure as a capacity ceiling)**.** *Let $I \in (0, 1]$ be infrastructure quality and $K_{\max}(I)$ the largest feasible capacity it admits. Then $K_{\max}$ is increasing in I; below a threshold I^* only $K = 1$ is feasible; and infrastructure enters the invariant through* three compounding channels*—it caps K, it raises $c_v(I)$, and it inflates travel time-variance, so that below a threshold I_S* no guaranteed window is promisable at any price*. The premium product does not become expensive; it ceases to exist.*

Theorem 4 (The substitution boundary)**.** *Digital infrastructure can substitute for the* information layer*—locatability records, coordination, payment—but never for the* mobility layer*—the physical movement of the unit. A ledger can replace a missing address system; it cannot replace a missing road.*

Corollary 4 (The portfolio rule)**.** *Digital investment has the highest return where the mobility layer already exists and only the information layer is missing. The leapfrog is real for information and impossible for mobility.*

9 The Diagnostic

The theory reduces practice to three observable numbers. They are computed, not estimated, from the bucket system's own records.

Proposition 2 (The utilisation diagnostic)**.** *For an achieved service rate R_{obs} and the theoretical rate $R(a)$ implied by locatability, define the* utilisation ratio $u = R_{obs}/R(a)$*. In steady state:*

$$u < 0.7 \text{ — } \textit{the system is } \text{load-constrained } \textit{(demand is the bottleneck);}$$

$$u > 0.9 \text{ — } \textit{the system is } \text{locatability-constrained } \textit{(search is the bottleneck).}$$

The first is answered by demand acquisition; the second by verification. The second number is the kill rate κ_W*—the fraction of buckets killed in a rolling window, whose healthy range is $\approx$ 0.10–0.25 and whose rise precedes the financials by weeks. The third is* units per unit time per unit capacity*, the speed of the loop.*

A system that pulls the wrong lever—verifying endpoints in a load-constrained system, or acquiring demand in a locatability-constrained one—invests in an asset that will not be the binding constraint when it matures. The diagnostic is the operational form of the theory, and it is the difference between a strategy and its absence.

10 What Would Refute It

A theory that cannot be refuted is a manifesto. The following observations would refute specific results, and each is measurable in the operation of any instance:

1. Against 1: a system is durably profitable whose departures never cross λ^*, with no subsidy from elsewhere.
2. Against 3: a purchased static-locatability moat that survives full public provision of the coordinate layer.
3. Against 1: a system below the separatrix rescued by capital alone, without a change in locatability or load factor.
4. Against 2: a capacity-constrained, deadline-committed service whose profitability is not governed by a load-factor threshold.
5. Against 3: a guaranteed window reliably promised where $I < I_S$.
6. Against 4: digital infrastructure making a physically impassable corridor passable.

11 Conclusion

The theory rests on one axiom and yields one law, one loop, and one principle. *Locate, then load*: make the endpoint calculable before you commit the capacity. The rate primitive says locatability is the only thing that moves the service rate. The consolidation invariant says a departure is worth firing only past a load-factor threshold set by fixed cost over capacity. The flywheel says the whole thing is a closed loop with two attractors and a separatrix at the break-even point. The durability principle says the only asset that survives is the one accumulated by operating—the event stream, not the coordinate. And the substitution boundary says what all of it cannot do: nothing digital can move a unit through a wall.

The address is not the asset. The address is the door.
The asset is what you learn about the room behind it, and how many times you have opened that door.

The remainder of this volume is commentary: each subsequent paper instantiates one theorem of this foundation in one domain, at one set of parameter values, to one operational end. They add no new primitives. They only fill the parameters in.